



Estimation of thermo-physical properties of products with cylindrical shape during drying: The coupling between mass and heat



Wilton Pereira da Silva^{*}, Cleide M.D.P.S. e Silva, Fernando J.A. Gama

Center of Sciences and Technology, Federal University of Campina Grande, PB, Brazil

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ABSTRACT

This article aimed at studying drying of products with cylindrical shape, using an apparent liquid diffusion model, with coupling between mass and heat at surface of the product. Model includes variable thermo-physical properties and shrinkage. The one-dimensional diffusion equation in cylindrical coordinates, with boundary condition of the third kind (i.e., convective boundary condition), was discretized by means of the finite volume method with a fully implicit formulation. The proposed solution can be used to determine thermo-physical parameters, via optimization technique, and also to simulate water migration and heat transport. Developed software was applied to describe convective drying of whole bananas. Three experiments were carried out at average drying air temperature of 47.9, 58.6 and 66.9 °C, enabling to determine the thermo-physical properties as well as to simulate the process of heat and mass transfer. The statistical indicators of the simulations made it possible to conclude that, for the three experiments, the proposed model well describes the involved processes.

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1. Introduction

Food processing usually involves a stage in which heat is transferred to, and/or removed from, the product (Da Silva et al., 2011). Preservation methods such as pasteurization and cooling, among others, involve heat transfer. Thus, to describe these and other processes, the thermal properties of the product should be known. If the range of temperature involved in the process is not very large, the thermal properties can be considered constant. Thus, this type of processing is generally easy to be described because, in many occasions, the mass of the product can be also considered constant along the time as, for instance, in cooling processes. In other types of processing, such as soaking, baking and drying, simultaneous heat and mass transfer between the product and the external medium are involved. In these cases, in order to describe such processes, the thermo-physical properties of the product must be also known. This type of processing is difficult to be described, particularly the heat transfer, due to the fact that the ratio water/dry matter in the product is continually changing along the time. In the specific case of drying, water is lost along the time, while the dry matter does not vary during this process. As a consequence, in this case, several properties involved in the description of the phenomenon

are variable, such as the density of the product, specific heat, thermal conductivity and thermal diffusivity (Lima et al., 2002; Mariani et al., 2008; Lemus-Mondaca et al., 2013; Perussello et al., 2013). Particularly in the drying study of fruits, the volume and moisture diffusivity of the product are also variable. As the drying operation is frequently used to prolong the shelf life of agricultural products, these properties must be known.

The post-harvest losses of agricultural products can be significantly reduced by using proper drying techniques. As above mentioned, the use of such techniques requires the knowledge of the thermo-physical properties of the agricultural product. Different approaches are used to determine thermo-physical properties and to describe water loss (and heat transfer) during drying, and some studies are presented in the following. In several drying studies, in order to describe heat and mass transfer using analytical solutions of the diffusion equation, some authors such as Ramsaroop and Persad (2012), which studied heat transfer during drying of coconut, assume the thermo-physical properties of the product with constant values. Wu et al. (2004) used numerical solutions of the diffusion equation to describe heat and mass transfer during drying of rice, but some thermo-physical properties of the product were considered constant by these researchers. However, most researchers consider these properties with variable values, and use numerical solutions of the diffusion equation to describe both: mass and heat transfer (Lima et al., 2002; Mariani et al., 2008; Lemus-Mondaca et al., 2013; Perussello et al., 2013). Usually the mass migration is described under isothermal

^{*} Corresponding author. Tel.: +55 83 3333 2962.

E-mail addresses: wiltonps@uol.com.br (W.P. da Silva), cleidedps@uol.com.br (C.M.D.P.S. e Silva), fernando_gama@uol.com.br (F.J.A. Gama).



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对圆柱形产品在干燥过程中热物理性质的估算：质量与热量之间的关联



威尔顿·佩雷拉·达席尔瓦, 克莱德·M.D.P.S·埃席尔瓦, 费尔南多·J.A·加马

科学与技术中心, 巴西帕拉伊巴冈德鲁邦大学, 帕拉伊巴冈德鲁州

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附加说明

本文旨在利用表面物质与质量之间耦合关系的表面扩散模型, 研究圆柱形产品的干燥过程。该模型考虑了可变的热物理性质以及收缩现象。在圆柱坐标系下, 采用一维扩散方程来描述过程, 并通过有限体积分法以完全隐式格式对边界条件进行离散处理。所提出的解决方案可用于通过优化技术确定热物理参数, 同时还可以用于模拟水分迁移和热量传递过程。开发的软件被应用于整个香蕉的干燥过程模拟, 在平均干燥空气温度为 47.9、58.6 和 66.9°C 的条件下进行了三次实验, 从而能够确定热物理性质, 并模拟热量和质量传递过程。通过统计指标的分析, 可以得出结论: 对于这三项实验而言, 所提出的模型能够很好地描述相关过程。

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1. 引言

食品加工通常包含一个将热量传递给产品或从产品中移除热量的阶段 (Da Silva 等, 2011 年)。诸如巴氏杀菌和冷却等保鲜方法都涉及热量的传递。因此, 为了描述这些过程以及其他过程, 必须了解产品的热性质。如果过程中涉及的温度范围不是很大, 那么产品的热性质可以视为常数。因此, 这种类型的加工通常很容易描述, 因为在许多情况下, 产品的质量也可以被视为随时间恒定的, 例如在冷却过程中。而在其他类型的加工过程中, 如浸泡、烘烤和干燥, 产品与外部介质之间会同时发生热量和质量的传递。在这种情况下, 为了描述这些过程, 必须了解产品的热物理性质。这种加工方式很难描述, 尤其是关于热量传递的部分, 因为产品中的水分与干物质的比例会随时间不断变化。在干燥过程中, 水分会随时间逐渐流失, 而干物质的质量则保持不变。因此, 在这种情况下……

在描述这一现象时, 有几个性质是变化的, 比如产品的密度、比热容、热导率以及热扩散率 (Lima 等人, 2002 年; Mariani 等人, 2008 年; Lemus-Mondaca 等人, 2013 年; Perussello 等人, 2013 年)。特别是在水果的干燥过程中, 产品的体积和水分扩散率也是可变的。由于干燥操作常被用来延长农产品的保质期, 因此这些性质是必须被考虑的。

通过使用适当的干燥技术, 可以显著减少农产品的产后损失。如前所述, 采用这些技术需要了解农产品的热物理特性。有多种方法可以用来确定热物理特性, 并描述干燥过程中的水分流失 (以及热量传递情况)。在一些研究报告中, 一些学者如 Ramsaroop 和 Persad (2012 年) 利用扩散方程的解析解来描述干燥过程中的热量和质量传递现象, 他们假设产品的热物理特性是恒定的。Wu 等人 (2004 年) 则使用扩散方程的数值解来描述大米干燥过程中的热量和质量传递现象, 但他们也认为产品的某些热物理特性是恒定的。Archer 们。不过, 大多数研究人员认为这些性质具有可变的值, 因此他们使用扩散方程的数值解来描述质量和热量的传递 (Lima 等人, 2002 年; Mariani 等人, 2008 年; Lemus-Mondaca 等人, 2013 年; Perussello 等人, 2013 年)。通常, 质量迁移是在等温条件下进行描述的。

✉ 通讯作者。电话: +55 83 3333 2962。

电子邮件地址: wiltonps@uol.com.br (W.P. da Silva), cleidedps@uol.com.br (C.M.D.P.S. e Silva), fernando_gama@uol.com.br (F.J.A. Gama)。

Nomenclature

A_w, A_p, A_e, B	coefficients of the discretized diffusion equation
a, b	fitting parameters
A_s	area of the surface of the cylinder (m^2)
c_p	specific heat of product ($\text{J kg}^{-1} \text{K}^{-1}$)
c_v	specific heat of vapor of water ($\text{J kg}^{-1} \text{K}^{-1}$)
D	effective moisture diffusivity ($\text{m}^2 \text{s}^{-1}$)
h_m	convective mass transfer coefficient (m s^{-1})
h_T	heat transfer coefficient (m s^{-1})
h_H	convective heat transfer coefficient h_H ($\text{W m}^{-2} \text{K}^{-1}$)
h_{fg}	latent heat of vaporization of water (J kg^{-1})
k	thermal conductivity ($\text{W m}^{-1} \text{K}^{-1}$)
L	height of cylinder (m)
M	local moisture content ($\text{kg}_{\text{water}} \text{kg}_{\text{dry matter}}^{-1}$)
$\bar{M}$	average moisture content ($\text{kg}_{\text{water}} \text{kg}_{\text{dry matter}}^{-1}$)
m	mass of the product
N	number of control volumes
N_p	number of experimental points
r	position inside the cylinder (m)
R	radius of the cylinder (m)
R^2	coefficient of determination
t	drying time (s)
T	temperature (K)
Δr	thickness of the control volumes (m)

Subscripts

0	initial
eq	equilibrium
f	final
m, T	moisture, thermal
w, p, e	west, nodal point, east

Greek symbols

α	thermal diffusivity ($\text{m}^2 \text{s}^{-1}$)
χ^2	chi-square
I^Φ	transport coefficient (dimension depends on the process under study)
Φ	dependent variable of the diffusion equation (dimension depends on the phenomenon under study)
λ	transport coefficient (dimension depends on the phenomenon under study)
ρ	density (kg m^{-3})
σ_i	standard deviation of the experimental moisture content at the point i

conditions, i.e., this phenomenon is described, for each drying air temperature, in an independent way of the distribution of temperature within the product. In addition, the water migration is considered to occur in the liquid state, and its vaporization occurs at the surface where, therefore, happens the coupling between mass and heat.

Some particular considerations are made in each research on the heat and mass transfer. As examples, some interesting works are mentioned in the following. Lima et al. (2002), describing heat and mass transfer during drying of bananas, considered the density multiplied by the specific heat, in the transient term of the diffusion equation, as a variable quantity. Mariani et al. (2008), describing the heat transfer during drying of bananas, considered this quantity with a constant value along time. On the other hand, models of Lima et al. (2002) and Mariani et al. (2008) include the energy due to heating of the vapor generated at temperature of the surface up to drying air temperature. Although some other works also include this energy, most researchers neglect it. A very common simplification is presented in the following. In the calculations, most researchers use the value of the latent heat of vaporization of free water instead the value this property in the product. Wu et al. (2004) further simplifies this problem by ignoring the latent heat of vaporization of water to describe heat transfer during rice drying. In a literature search, the authors of the present paper did not find a study evaluating the implications of these simplifications above mentioned. In this sense, the aim of this article is defined in the following.

Basically, this article had two distinct objectives. First, using experimental data sets and the inverse method, a liquid diffusion model was proposed to describe heat and mass transfer during drying of whole bananas, determining their thermo-physical properties. For this purpose, the diffusion equation was numerically solved, by means of the finite volume method, with a fully implicit formulation. This solution, as well as the determination of the thermo-physical properties, permits to simulate the migration of both: mass and heat. The second objective of this article was to evaluate some simplifications proposed by researchers in the description of the heat transfer during drying.

2. Materials and methods

2.1. Drying experiments

Mature bananas *Musa acuminata*, subgroup Cavendish cv nanica was acquired from the local market, Campina Grande, Brazil. The bananas were peeled and selected by their appearance, with no evidence of mechanical damage. The experiments were carried out in a convective dryer with vertical flux, controller of temperature and controller of air velocity (Fig. 1). One sieve (number 1) with one whole banana was placed in the convective dryer, with hot air at the following average temperatures: 47.9, 58.6 and 66.9 °C. In each experiment, the drying air temperature was measured twenty times along the process, and the correspondent standard deviations for the average values of 47.9, 58.6 and 66.9 °C were 0.7, 0.8 and 0.6 °C, respectively. During the experiments, moisture content was measured by the gravimetric method. The bananas were weighed at time intervals ranging from 5 min at the beginning of drying to about 4 h at the final part of the process. At the end of each drying, the banana was removed from the dryer and placed within an oven at 105 °C. The product remained there

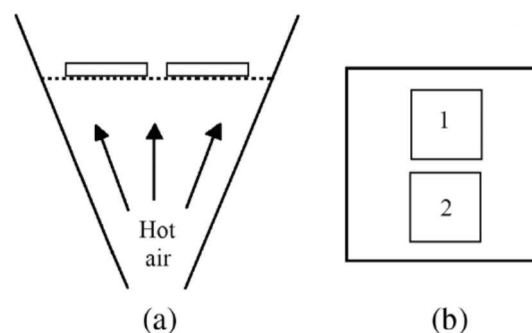


Fig. 1. Convective dryer: (a) Cross-section; (b) positions of the trays on the cross-section: number 1 – banana for mass measurement and number 2 – banana with thermocouple for temperature measurement.

命名法	
离散化扩散方程的 A、A、A，以及 B 系数	
a、b 拟合参数	
圆柱体表面的某一部分面积（单位：米）	
c 产品的比热容（单位：J/kg·K）	
水的比热容（单位：焦耳/千克·开）	
D 有效水分扩散系数（ms）	
h 对流传质系数（m s）	
热传递系数（单位：米秒）	
h 对流传热系数，单位：W·mK	
水的潜热（单位：焦耳每千克）	
k 热导率（单位：W·mK）	
L 圆柱体高度（米）	
M 局部水分含量（以干物质计，单位：千克每千克）	
M 平均水分含量（单位：千克每千克干物质）	
m 产品的质量	
N 控制体积的数量	
N 实验得分数量	
r 在圆柱体内的位置（单位：米）	
R 圆柱体的半径（单位：米）	
R 决定系数	
t 干燥时间（秒）	
T 温度（开尔文）	
博士 控制体积的厚度（单位：米）	
下标	
0 初始值	
均衡	
f 最终版/完成版	
m, T 湿度，温度 w, p, e 能量	
西方，中心点，东方	
希腊符号	
a	热扩散率（ms）
v	卡方检验
C	传输系数（其维度取决于所研究的过程）
U 扩散方程的因变量（其维度取决于所研究的现象）	
k 传输系数（其维度取决于所研究的物理现象）	
正在研究的名词）	
q	密度（千克/米 ³ ）
r	实验测得的含水率的标准偏差，即在该点处的偏差

在这种情况下，这种现象是针对每一种干燥空气温度而单独描述的，且考虑了产品内部温度分布的不同情况。此外，水的迁移被认为是在液态下发生的，而其蒸发则发生在表面，因此，质量和热量之间的耦合关系也就体现在了表面上。

在每项关于热传递和质量传递的研究中，都会考虑一些特定的因素。例如，以下提到了一些有趣的研究。Lima 等人（2002 年）在研究香蕉的干燥过程中热传递和质量传递时，将扩散方程中的密度乘以比热作为变量来处理。Mariani 等人（2008 年）在描述香蕉干燥过程中的热传递时，认为这一数值随时间保持不变。而 Lima 等人（2002 年）和 Mariani 等人（2008 年）的模型则考虑了表面温度下的蒸汽加热所释放的能量，直到空气温度达到该温度为止。虽然其他一些研究也考虑了这种能量，但大多数研究者都忽略了它。下面给出了一种常见的简化处理方法：在计算中，大多数研究者使用自由水的汽化热值，而不是其他数值。在产品中存在这种性质。Wu 等人（2004 年）通过忽略水的汽化潜热来进一步简化这个问题，从而描述水稻干燥过程中的热传递现象。在文献检索中，本文作者并未找到任何研究来评估上述简化方法的实际意义。因此，本文的研究目的可以定义为：探讨这种简化方法在实际应用中的可行性。

这篇文章主要有两个目的。首先，利用实验数据以及反分析方法，建立了一个液体扩散模型，用于描述整根香蕉在干燥过程中热量和质量的传递情况，从而确定其热物理特性。为此，采用了有限体积法对扩散方程进行数值求解，该方法的公式完全采用隐式形式表示。这种求解方法，再加上对热物理特性的确定，使得能够同时模拟质量和热量的传递过程。本文的第二个目的是评估研究人员在描述干燥过程中热量传递时提出的一些简化方法的有效性。

2. 材料与方法

2.1 干燥实验

从巴西 Campina Grande 地区的当地市场采购了成熟的香蕉，品种为 *Musa acuminata*，属于 Cavendish 亚组，品种名称为 nanica。这些香蕉经过去皮后，根据外观进行挑选，没有发现机械损坏的迹象。实验在一种具有垂直气流的干燥机中进行，该干燥机还配备了温度控制器和风速控制器（见图 1）。将一个完整的香蕉放在对流干燥机中，同时向其中通入不同温度的热空气：47.9°C、58.6°C 和 66.9°C。在每次实验中，沿着干燥过程对空气温度进行了二十次测量，47.9°C、58.6°C 和 66.9°C 时的平均温度标准偏差分别为 0.7°C、0.8°C 和 0.6°C。在实验过程中，通过重量法测量了香蕉的含水量。香蕉在干燥过程中的不同时间被称重，从干燥开始时的 5 分钟到最终阶段的约 4 小时不等。每次干燥结束后，香蕉会被取出并放入温度为 105°C 的烤箱中加热。经过这样的处理之后，产品就完成了加工过程。

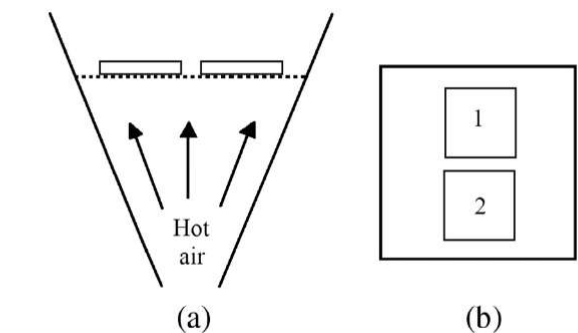


图 1. 对流干燥机：(a) 横截面图；(b) 横截面上各个托盘的位置：编号 1 的托盘用于测量质量，而编号 2 的托盘则配备了热电偶，用于测量温度。

Table 1

Temperature, dimension, moisture content, dry matter and specific heat of vapor for each drying air temperature.

T_{air} (°C)	L_0 (mm)	R_0 (mm)	M_0 (db)	M_f (db)	M_{eq} (db)	m_d (kg)	c_v (J kg ⁻¹ °C ⁻¹)
47.9	141	14.9	3.43	0.202	0.121	0.0192	1908
58.6	143	14.8	3.14	0.224	0.108	0.0223	1911
66.9	165	15.3	2.90	0.228	0.094	0.0276	1914

for 24 h, enabling the measurement of the dry matter. The average drying air temperatures, dimensions, moisture contents, dry matters and specific heats of the vapor are given in Table 1.

Information about the drying air (average relative humidity, $\overline{RH}$, and velocity, $\bar{v}$) and the room air (average temperature and relative humidity) is given in Table 2 for each drying air temperature. The relative humidity, velocity of the air drying as well as room temperature was measured twenty times along of each experiment, and the values provided in Table 2 are the average values obtained in each measurement process.

The radius R and height L was measured every instant in which the bananas were removed from the dryer to measure the mass. This way, the ratio between the current and initial radius, R/R_0 , was determined at every instant. Expressions R/R_0 as a function of the average moisture content $\bar{M}$ were obtained using curve fitting for all drying air temperatures, as is shown in Table 3.

If the expressions R/R_0 (Table 3) are drawn as a function of the average moisture content for all drying air temperatures in a same system of axes, the obtained lines are very close one of each other. On the other hand, the measurements of mass, radius and height enabled to determine the density of the bananas along the drying process. Thus, it is possible to determine an expression for the density of the bananas as a function of the moisture content of the product.

A thermocouple was placed in the center of a second banana (sieve number 2), which was also placed in the dryer. The initial temperature of the banana is the room temperature, which is provided in Table 2. For each drying air temperature T , care was exercised to select two bananas with the same dimensions (or very close) to minimize size effect on the experimental data. The data-sets of temperature were collected in regular time interval by using a digital Multimeter Cole Parmer, with temperature range of -180° to 370°C ; and accuracy of 0.1°C in the temperature range of interest.

2.2. One-dimensional diffusion equation and its discretization

In order to describe the processes involved during drying, the following assumptions were considered: (1) the product was considered homogeneous and isotropic; (2) the water migration can be described by liquid diffusion model; (3) the mass migration is considered under isothermal conditions, since at initial instants of drying, the thermal diffusivity ($10^{-7}\text{ m}^2\text{ s}^{-1}$) is greater than the moisture diffusivity ($10^{-10}\text{ m}^2\text{ s}^{-1}$) and, consequently, the Biot number for heat transfer is lesser than the mass transfer Biot number; (4) the mass and thermal diffusivities are variable properties; (5) the coupling between mass and heat occurs at surface, where the liquid phase is transformed to vapor. In addition to

Table 2

Relative humidity, velocity and temperature.

Drying air			Room air	
T (°C)	$\overline{RH}$ (%)	$\bar{v}$ (m/s)	T_0 (°C)	$\overline{RH}_{room}$ (%)
47.9	20.3	1.84	25.3	59.3
58.6	10.4	1.70	25.6	48.3
66.9	6.6	1.50	24.6	48.4

Table 3Ratio R/R_0 as a function of the average moisture content.

T (°C)	R/R_0
47.9	$(0.2500\bar{M} + 0.1576)^{1/3}$
58.6	$(0.2821\bar{M} + 0.1123)^{1/3}$
66.9	$(0.3016\bar{M} + 0.1160)^{1/3}$

these assumptions, it was established that radial shrinkage must be included to model.

For a cylinder of radius R and height L , in which $L \gg R$, the one-dimensional diffusion equation can be written in the following way:

$$\frac{\partial(\lambda\Phi)}{\partial t} = \frac{1}{r} \frac{\partial}{\partial r} \left(r \Gamma^\Phi \frac{\partial \Phi}{\partial r} \right) \quad (1)$$

where Φ is the variable that describes a diffusive process (its unit depends on the specific process); t is the time (s); r defines the radial position within the cylinder (m); λ and Γ^Φ are the properties of the product (their units depend on the specific process). In order to numerically solve Eq. (1) using the finite volume method with fully implicit formulation (Patankar, 1980), the domain was divided into N control volumes, generating a uniform grid in the circular section of the cylinder, as is shown in Fig. 2a.

Fig. 2b shows details of an internal control volume P and its neighbors to west (W) and east (E). Integrating Eq. (1) on space, $2\pi r_p \Delta r L$, and time, from t to $t + \Delta t$, the following result is obtained:

$$\frac{\lambda_p \Phi_p - \lambda_p^0 \Phi_p^0}{\Delta t} r_p \Delta r = r_e \Gamma_e^\Phi \frac{\partial \Phi}{\partial r} \Big|_e - r_w \Gamma_w^\Phi \frac{\partial \Phi}{\partial r} \Big|_w \quad (2)$$

in which the subscripts p , e and w represent nodal point; interface east and interface west, respectively. The term with superscript 0 is

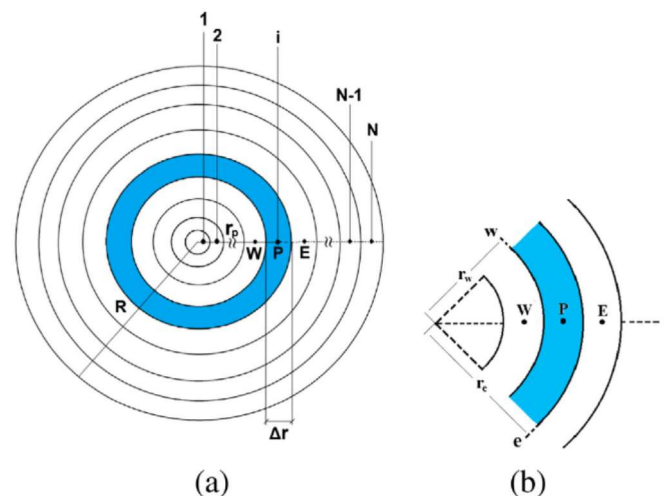


Fig. 2. (a) Uniform grid in the circular section of the infinite cylinder highlighting an internal control volume with nodal point P ; (b) details of the control volume P and its neighbors to west (W) and east (E).

表 1

每种干燥空气温度下的温度、尺寸、水分含量、干物质含量以及汽化热。

温度 (摄氏度)	L(毫米)	R(毫米)	M(db)	M (数据库)	M(db)	m(千克)	摄氏度 (J 千克度)
47.9	141	14.9	3.43	0.202	0.121	0.0192	1908
58.6	143	14.8	3.14	0.224	0.108	0.0223	1911
66.9	165	15.3	2.90	0.228	0.094	0.0276	1914

在 24 小时内, 可以测量干物质含量。各试验点的干燥空气温度、尺寸、水分含量、干物质含量以及蒸汽的比热容等参数均记录在表 1 中。

关于干燥空气的信息 (平均相对湿度、风速以及总速度), 以及房间内的空气情况 (平均温度和相对湿度), 都在表 2 中有所说明。在每次实验过程中, 空气干燥的相对湿度、风速以及房间温度分别被测量了 20 次, 表 2 中给出的数值是每次测量过程得到的平均值。

在每次取出香蕉并测量其质量时, 都会同时测量香蕉的半径 R 和高度 L。通过这种方式, 可以计算出当前半径与初始半径的比值 R/R₀。通过对所有干燥空气温度进行曲线拟合, 得到了与平均水分含量 M 相关的 R/R₀ 的表达式, 具体结果见表 3。如果将表达式 R/R₀ (表 3) 表示为同一系统中所有干燥空气温度下的平均水分含量的函数, 那么得到的曲线彼此非常接近。另一方面, 通过测量香蕉的质量、半径和高度, 可以确定在干燥过程中香蕉的密度。因此, 可以推导出香蕉密度与产品水分含量之间的关系表达式。

在第二个香蕉样本 (筛孔数为 2) 的中心位置放置了一个热电偶, 该样本也被放置在干燥机中。香蕉的初始温度即为室温, 具体数值可参见表 2。对于每一个干燥空气温度 T, 都精心选择了两个尺寸相同 (或非常接近) 的香蕉样本, 以尽量减少尺寸因素对实验数据的影响。温度数据是通过使用 Cole Parmer 品牌的数字万用表定期采集的, 其温度测量范围可达 180°C 至 370°C, 精度在 0.1°C 以内。

2.2 一维扩散方程及其离散化 为了描述干燥过程中发生的现象, 考虑了以下假设:

(1) 被处理的物质被认为是均匀且各向同性的; (2) 水分迁移可以用液体扩散模型来描述; (3) 质量迁移发生在恒温条件下, 因为在干燥初期, 热扩散系数 (10ms) 大于水分扩散系数 (10ms), 因此, 传热的比奥数小于质量传递的比奥数; (4) 质量扩散系数和热扩散系数是可变的属性; (5) 质量和热量之间的耦合发生在表面, 此时液相转化为蒸汽。

表 2

相对湿度、风速和温度。

干燥的空气			室内的空气	
温度 (°C)	RH (%)	v	温度 (摄氏度)	RH (%)
47.9	20.3	1.84	25.3	59.3
58.6	10.4	1.70	25.6	48.3
66.9	6.6	1.50	24.6	48.4

Table 3

Ratio R/R₀ as a function of the average moisture content.

C

T (°C)	R/R ₀
47.9	0.2500M ± 0.1576p
58.6	0.2821M ± 0.1123p
66.9	0.3016M ± 0.1160p

除了这些假设之外, 还确定必须考虑径向收缩因素来进行模型构建。

对于一个半径为 R、高度为 L 的圆柱体, 其中 L 远大于 R, 一维扩散方程可以表示为如下形式:

$$\frac{\partial k}{\partial t} U \frac{1}{r} \frac{\partial}{\partial r} r C = \frac{\partial}{\partial r} r C \quad (1)$$

其中, U 是描述扩散过程的变量 (其单位取决于具体的过程); t 是时间 (单位: 秒); r 表示圆柱体内的径向位置 (单位: 米); k 和 Care 则是该产品的属性参数 (它们的单位也取决于具体的过程)。为了使用有限体积法并采用完全隐式格式来数值求解方程 (1) (Patankar, 1980), 需要将域划分为 N 个控制体积, 从而在圆柱的圆形部分形成一个均匀的网格结构, 如图 2a 所示。

图 2b 展示了内部控制容积 P 及其相邻区域在西侧 (W) 和东侧 (E) 的详细信息。将方程(1)在空间、2pD 和时间上进行积分, 从 D 开始计算, 可以得到以下结果:

$$\frac{k U}{D t} - R \frac{\partial}{\partial r} r C = \frac{\partial}{\partial r} r C \quad (2)$$

其中, 下标 p、e 和 w 分别代表节点; 分别对应于东界面和西界面。带有上标 0 的术语则是……

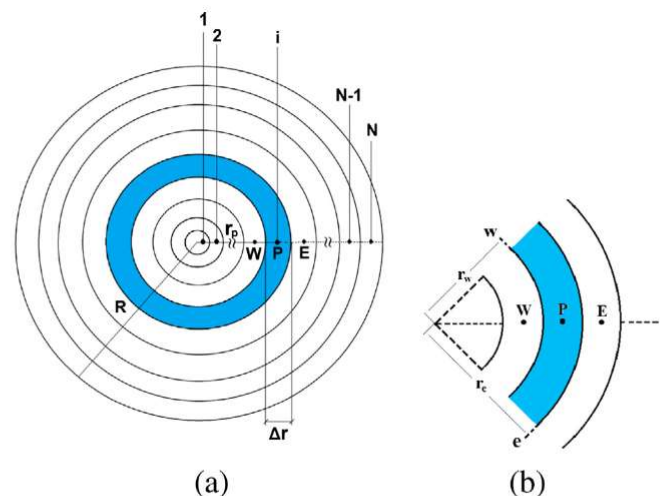


Fig. 2. (a) Uniform grid in the circular section of the infinite cylinder highlighting an internal control volume with nodal point P; (b) details of the control volume P and its neighbors to west (W) and east (E).

calculated at beginning of the time interval (t), while the terms without this superscript are calculated at the end ($t + \Delta t$). Eq. (2) is a partial result that will be used to obtain algebraic equations for each type of control volume as is shown in the following.

2.2.1. Internal control volumes

For this type of control volume, with neighbors to west and east, the derivatives of Eq. (2) can be given in the following way:

$$\left. \frac{\partial \Phi}{\partial r} \right|_e \cong \frac{\Phi_E - \Phi_P}{\Delta r} \quad \left. \frac{\partial \Phi}{\partial r} \right|_w \cong \frac{\Phi_P - \Phi_W}{\Delta r} \quad (3a-b)$$

Substituting Eq. (3a-b) into Eq. (2), the following algebraic equation is obtained:

$$A_P \Phi_P + A_W \Phi_W + A_E \Phi_E = B \quad (4)$$

in which

$$A_W = -\frac{r_w}{\Delta r} \frac{\Gamma_w^\Phi}{\lambda_p} \quad A_P = \frac{r_P \Delta r}{\Delta t} + \frac{r_e}{\Delta r} \frac{\Gamma_e^\Phi}{\lambda_p} + \frac{r_w}{\Delta r} \frac{\Gamma_w^\Phi}{\lambda_p} \quad (5a-d)$$

$$A_E = -\frac{r_e}{\Delta r} \frac{\Gamma_e^\Phi}{\lambda_p} \quad B = \frac{\lambda_P^0 r_P \Delta r}{\lambda_p \Delta t} \Phi_P^0$$

2.2.2. Control volume number 1

For the control volume at the center of the grid, due to the symmetry, the west mass flux is zero. Consequently, the derivative given in Eq. (3b) is zero. Thus, for control volume number 1 it is obtained:

$$A_P \Phi_P + A_E \Phi_E = B \quad (6)$$

where

$$A_P = \frac{r_P \Delta r}{\Delta t} + \frac{r_e}{\Delta r} \frac{\Gamma_e^\Phi}{\lambda_p} \quad A_E = -\frac{r_e}{\Delta r} \frac{\Gamma_e^\Phi}{\lambda_p} \quad B = \frac{\lambda_P^0 r_P \Delta r}{\lambda_p \Delta t} \Phi_P^0 \quad (7a-c)$$

2.2.3. Water removal and heat transport

In order to apply the above results in the water removal, in Eq. (1) it should be imposed: $\Phi \equiv M$, in which M is the moisture content in dry basis ($\text{kg}_{\text{water}}/\text{kg}_{\text{dry matter}}$); $\lambda = 1$ (dimensionless); $\Gamma^\Phi = D$, i.e., effective moisture diffusivity ($\text{m}^2 \text{s}^{-1}$).

As an observation, in order to describe heat transport, the thermal properties referring to advection, radiation, phase change, heat sources and conduction should be known. Many times, solely the conduction mechanism and phase change are used to describe heat transfer and, consequently, the properties involved (such as thermal diffusivity and conductivity) are considered as “apparent”. However, in this article, for simplicity, the thermal diffusivity will continue to be named with this same name.

In the heat conduction study, the imposition on the variables of Eq. (1) Φ , λ and Γ^Φ are as follows: $\Phi \equiv T$, i.e., temperature (K); $\lambda = \rho c_p$, where ρ is the density (kg m^{-3}) and c_p is the specific heat of the product ($\text{J kg}^{-1} \text{K}^{-1}$); $\Gamma^\Phi = k$, i.e., thermal conductivity ($\text{W m}^{-1} \text{K}^{-1}$). If the thermal conductivity is divided by ρc_p , a thermal property called thermal diffusivity ($\text{m}^2 \text{s}^{-1}$) is obtained: $\alpha = k/\rho c_p$.

For the control volume N , which is in contact with the external medium, two different equations must be obtained to complete the discretization of the diffusion equation to describe mass and heat transfer. These equations are obtained in the following.

2.2.4. Water removal: control volume N at boundary of the cylinder

For the control volume N , supposing boundary condition of the third kind, the following discretized equation is obtained:

$$A_P M_P + A_W M_W = B \quad (8)$$

where

$$A_P = \frac{r_P \Delta r}{\Delta t} + \frac{r_e D_e}{h_m + \frac{\Delta r}{2}} + \frac{r_w D_w}{\Delta r} \quad A_W = -\frac{r_w D_w}{\Delta r} \quad B = \frac{r_P \Delta r}{\Delta t} M_P^0 + \frac{r_e D_e}{h_m + \frac{\Delta r}{2}} M_{eq} \quad (9a-c)$$

in which h_m is the convective mass transfer coefficient (m s^{-1}) and M_{eq} is the equilibrium moisture content (db). Detailed steps on the obtaining of Eq. (9a-c) for moisture migration are presented by Silva et al. (2012a).

2.2.5. Heat transport: control volume N at boundary of the cylinder

In this article, apparent liquid diffusion was assumed. Thus, liquid water migrates from internal medium to the surface of the product, where occurs change to vapor. Thus, for the control volume N , the boundary condition of the third kind is expressed in the following way:

$$-\alpha_e \left. \frac{\partial T}{\partial r} \right|_e = h_T (T_e - T_{eq}) - \frac{[h_{fg} + c_v (T_e - T_{eq})] \frac{dm}{dt}}{\rho c_p A_s} \quad (10)$$

where h_T is the heat transfer coefficient (m s^{-1}); T_{eq} is the equilibrium temperature (K); c_v is specific heat of the vapor ($\text{J kg}^{-1} \text{K}^{-1}$); h_{fg} is the latent heat of vaporization (J kg^{-1}); dm/dt is the water loss rate (kg s^{-1}) and A_s is the area of the surface of the cylinder (m^2), given by $A_s = 2\pi RL$. In this article, it was assumed that the ratios L/L_0 are given by the same expressions provided in Table 3 for the radius. The convective heat transfer coefficient h_H ($\text{W m}^{-2} \text{K}^{-1}$) is related with h_T through the expression $h_T = h_H/(\rho c_p)$. On the other hand, combining the result to be obtained with the discretization of Eq. (10) with Eq. (2) applied to heat transfer, Eq. (11) is determined:

$$A_P T_P + A_W T_W = B \quad (11)$$

in which

$$A_P = \frac{r_P \Delta r}{\Delta t} + \frac{\alpha_e r_e}{h_T - \frac{c_v}{\rho c_p A_s} \frac{dm}{dt} + \frac{\Delta r}{2}} + \frac{r_w}{\Delta r} \alpha_w \quad A_W = -\frac{r_w}{\Delta r} \alpha_w$$

$$B = \frac{\rho^0 c_p^0 r_P \Delta r}{\rho c_p \Delta t} T_P^0 + \frac{\alpha_e r_e}{h_T - \frac{c_v}{\rho c_p A_s} \frac{dm}{dt} + \frac{\Delta r}{2}} T_{eq} + \frac{\alpha_e r_e \frac{h_{fg}}{\rho c_p A_s} \frac{dm}{dt}}{\alpha_e + \frac{\Delta r}{2} \left(h_T - \frac{c_v}{\rho c_p A_s} \frac{dm}{dt} \right)} \quad (12a-c)$$

2.3. Mass and thermal diffusivity

For the nodal points, the diffusivity Γ^Φ (D or α) can be calculated from an appropriate relation between such property and the value of Φ ,

$$\Gamma^\Phi = f(a, b, \Phi) \quad (13)$$

where a and b are parameters that fit the numerical solution to the experimental data, and they can be determined by optimization.

On the interfaces of the internal control volumes, for example e , for uniform grids the following expression (harmonic mean) should be used to determine Γ^Φ (Patankar, 1980):

$$\Gamma_e^\Phi = \frac{2\Gamma_e^\Phi \Gamma_P^\Phi}{\Gamma_e^\Phi + \Gamma_P^\Phi} \quad (14)$$

If the diffusivity is constant, the coefficients A of the discretized equations are calculated only once, and B is calculated in each time step because its value depends on Φ_P^0 , which is the value of Φ in the control volume P at the initial instant of each time step. On the other hand, if Γ^Φ is variable, the coefficients A are also calculated in each time step, due to the nonlinearities caused by the variation of such parameter. In this case, if the time refinement is adequate, errors due to the nonlinearities can be discarded.

在时间间隔 t 开始时进行计算，而没有任何上标的项则是在时间 t+D_t 时进行计算的。方程(2)是一个部分结果，它将用于得到每种控制体积对应的代数方程，具体如下所示。

2.2.1 内部控制体积 对于这种类型的控制体积，由于其西侧和东侧都有邻近区域，因此方程(2)的衍生物可以如下表示：

$$\frac{\partial U}{\partial t} = \frac{U}{\Delta x} \left(\frac{\partial U}{\partial x} \right)_{\text{博士}} - \frac{\partial U}{\partial x} \left(\frac{\partial U}{\partial x} \right)_{\text{博士}} \quad \delta 3a-bP$$

将方程(3a-b)代入方程(2)中后，可以得到以下代数方程：

$$A \frac{\partial U}{\partial t} = \frac{\partial U}{\partial x} \left(\frac{\partial U}{\partial x} \right)_{\text{博士}} \quad \delta 4P$$

在其中

$$\frac{\partial U}{\partial t} = \frac{\partial U}{\partial x} \left(\frac{\partial U}{\partial x} \right)_{\text{博士}} - \frac{\partial U}{\partial x} \left(\frac{\partial U}{\partial x} \right)_{\text{博士}} \quad \delta 5a-dP$$

2.2.2. 控制体积 1 的计算 对于位于网格中心的那个控制体积，由于对称性，西向的质量通量为零。因此，方程(3b)中给出的导数也为零。所以，对于控制体积 1，可以得到如下结果：

$$A \frac{\partial U}{\partial t} = \frac{\partial U}{\partial x} \left(\frac{\partial U}{\partial x} \right)_{\text{博士}} \quad \delta 6P$$

在何处

$$\frac{\partial U}{\partial t} = \frac{\partial U}{\partial x} \left(\frac{\partial U}{\partial x} \right)_{\text{博士}} - \frac{\partial U}{\partial x} \left(\frac{\partial U}{\partial x} \right)_{\text{博士}} \quad \delta 7a-cP$$

2.2.3 水分去除与热量传输 为了将上述结果应用于水分去除过程，在方程(1)中应当引入以下表达式：U = M，其中 M 表示干基水分含量（单位：千克/千克干物质）；k=1（无量纲）；C=D,i，即有效水分扩散系数（ms）。

作为一个观察结果，为了描述热传递过程，需要了解与对流、辐射、相变、热源以及传导相关的一系列热性质。很多时候，人们仅使用传导机制和相变来描述热传递过程，因此相关的性质（如热扩散率和导热性）被视作“隐性的”。不过，在本文中，为了简便起见，这些性质仍将继续使用相同的名称来表示。

在热传导研究中，方程(1)中的变量 U、k 和 C 的赋值如下：U = T，即温度（单位：K）；k = QQ，即密度（单位：千克每立方米）；C = k，即热导率（单位：瓦特每米开尔文）。如果以热导率除以 q，则得到热扩散率（单位：米秒）：α = q/T。

对于与外部介质接触的控制体积 N，需要得到两个不同的方程，以完成扩散方程的离散化，从而描述质量传递和热量传递的过程。这些方程如下所示。

2.2.4. 水分去除：在圆柱体的边界处控制体积 N。对于控制体积 N，假设边界条件为第三种类型，则可以得到以下离散化方程：

$$A \frac{\partial M}{\partial t} = \frac{\partial M}{\partial x} \left(\frac{\partial M}{\partial x} \right)_{\text{博士}} \quad \delta 8P$$

在何处

$$\frac{\partial U}{\partial t} = \frac{\partial U}{\partial x} \left(\frac{\partial U}{\partial x} \right)_{\text{博士}} - \frac{\partial U}{\partial x} \left(\frac{\partial U}{\partial x} \right)_{\text{博士}} \quad \delta 9a-cP$$

其中，他推导出了对流质量传递系数（m_s）和平衡水分含量（db）。Silva 等人（2012a）详细介绍了获取方程（9a-c）以描述水分迁移过程的步骤。

2.2.5. 热传输：在圆柱体的边界上控制体积 N 在本文中，假设了表观液体扩散现象。因此，液态水会从内部介质迁移到产品表面，并在那里转化为蒸汽。因此，对于控制体积 N 来说，第三种类型的边界条件可以如下表述：

$$\frac{\partial U}{\partial t} = \frac{\partial U}{\partial x} \left(\frac{\partial U}{\partial x} \right)_{\text{博士}} - \frac{\partial U}{\partial x} \left(\frac{\partial U}{\partial x} \right)_{\text{博士}} \quad \delta 10P$$

其中，传热系数表示为 m_s；T 为平衡温度，单位为 K；c 为蒸汽的比热，单位为 J/kgK；l 为汽化潜热，单位为 J/kg；dm/dt 为水的流失率，单位为 kg/s；A 为圆柱表面的面积，计算公式为 A=2p。在本文中，假设 L/L 的比值由表 3 中给出的与半径相关的表达式确定。对流传热系数 h(W/mK)可以通过 h=h/(q)来表示。将方程(10)的离散化结果与方程(2)结合起来，就可以得到方程(11)的解。

$$AT \frac{\partial T}{\partial t} = \frac{\partial T}{\partial x} \left(\frac{\partial T}{\partial x} \right)_{\text{博士}} \quad \delta 11P$$

在其中

$$\frac{\partial T}{\partial t} = \frac{\partial T}{\partial x} \left(\frac{\partial T}{\partial x} \right)_{\text{博士}} - \frac{\partial T}{\partial x} \left(\frac{\partial T}{\partial x} \right)_{\text{博士}} \quad \delta 12a-cP$$

2.3. 质量和热扩散率 对于节点 n，扩散率 C(D 或)

这种财产与 U 的价值之间存在着恰当的关系。

$$C \frac{\partial f}{\partial t} = \frac{\partial f}{\partial x} \left(\frac{\partial f}{\partial x} \right)_{\text{博士}} \quad \delta 13P$$

其中，a 和 b 是用于拟合实验数据数值解的参数，这些参数可以通过优化方法来确定。

例如，在内部控制体积的界面上，对于均匀网格，应使用以下表达式（潜波平均值）来计算 C(Patankar, 1980)：

$$C \frac{\partial f}{\partial t} = \frac{\partial f}{\partial x} \left(\frac{\partial f}{\partial x} \right)_{\text{博士}} \quad \delta 14P$$

如果扩散系数保持恒定，那么离散化方程中的系数 A 只需计算一次即可。而系数 B 则需要每个时间步长中重新计算，因为其值取决于 U 的值。而 U 的值则是在每个时间步长的初始时刻控制体积 P 中的 U 值。另一方面，如果 C 是变量，由于该参数的变化会导致非线性效应，因此系数 A 也需要在每个时间步长中重新计算。在这种情况下，如果时间细化足够精确，那么由非线性效应引起的误差可以被忽略不计。

In this article, the function $f(a, b, M)$ used to relate the effective mass diffusivity with the local value of the moisture content is given by (Silva et al., 2012a):

$$D = b_m \exp(a_m M), \quad (15)$$

where a_m and b_m are parameters to be determined by optimization and M is the local moisture content in dry basis.

It is well known that the thermal diffusivity of a product depends on its composition. In this sense, the thermal diffusivity of fresh fruits is very high, due to their high moisture content. During drying, the thermal diffusivity of a fruit at any instant depends on its water quantity. As water is continually lost during drying, i.e., the moisture content of the product is diminishing, the value of the thermal diffusivity also diminishes along time (Lima et al., 2002; Mariani et al., 2008). Thus, in this article, for each drying air temperature, the following expression was proposed for this property:

$$\alpha = b_T \exp(a_T \bar{M}^2), \quad (16)$$

in which a_T and b_T are constants for each drying air temperature, and are determined by optimization. On the other hand, $\bar{M}$ is the average moisture content of the fruit at any instant during the drying process.

2.4. General considerations

The water loss rate, dm/dt , necessary to describe heat transport, can be determined when water migration is described. Thus, initially water migration was studied. In each time step, a system of equations (Eqs. (4), (6), and (8)) is solved using Tridiagonal Matrix Algorithm method, TDMA (Press et al., 1996). Since water migration is described, in each time step the water loss rate is determined by:

$$\frac{dm}{dt} = \frac{m(t + \Delta t) - m(t)}{\Delta t} \quad (17)$$

in which the mass of water (kg) at instant t is given by $m(t) = m_d \bar{M}(t)$ where m_d is the mass of the dry matter (kg), given in Table 1, and $\bar{M}(t)$ is the average moisture content in dry basis ($\text{kg}_{\text{water}} \text{kg}_{\text{dry matter}}^{-1}$), calculated by:

$$\bar{M}(t) = \frac{\sum_{i=1}^N M_i r_i}{\sum_{i=1}^N r_i} \quad (18)$$

where M_i is the moisture content in the control volume “ i ” and r_i is its radius (m). Due to the shrinkage, in each time step the radius of the cylinder varies and, therefore, its value must be recalculated, as well as the thickness Δr of the control volumes of the uniform grid. Expressions for these calculations are given in Table 3.

The latent heat of water vaporization at the surface of the bananas was calculated according to the expression provided by Silva et al. (2012b):

$$h_{fg} = 2529.1 M^{(-1.782 \times 10^{-2} - 3.570 \times 10^{-4} T)} - 2.386 T \quad (19)$$

where h_{fg} is obtained in kJ kg^{-1} when M is given in dry basis and T in $^{\circ}\text{C}$.

For a given time step, in order to use Eqs. (4), (6), and (11) to determine the temperature in each control volume, the specific heat and density must be known at beginning and final of each time step. This is obvious because the coefficients B of these equations depend on these values. The specific heat of the bananas was estimated according to the expression provided by Sweat (1986) for agricultural products:

$$c_p = 1381 + 2930 X \quad (20)$$

where X is the average moisture content, in wet basis (decimal), and c_p is obtained in $\text{J kg}^{-1} \text{K}^{-1}$.

Specific heat for the water vapor, c_v , is provided in Table 1. On the other hand, with the measurements accomplished in the experiments, it was possible to determine an expression for the density ρ of the bananas (kg m^{-3}) as a function of the average moisture content, $\bar{M}$ (db), by curve fit:

$$\rho = 1533 e^{-0.1704 \bar{M}} \quad (21)$$

Thus, in each time step, if the values of the thermal diffusivity and heat transfer coefficient are provided in Eqs. (4), (6), and (11), the obtained system can be solved to determine the temperature in each control volume. At each time step, the system of equations was solved by using of TDMA method. Similar comment is also valid to study mass migration. Note that, if the thermal diffusivity and heat transfer coefficient are unknown, the parameters a_T , b_T and h_T can be determined by optimization (inverse method) using experimental data set, as is presented in the following.

2.5. Optimization

In order to determine the parameters I^{Φ} (D or α) and h (h_m or h_T), the parameters a and b of the function $I^{\Phi} = f(a, b, \Phi)$ and h can be determined by optimization using an experimental dataset. The expression of the chi-square (Bevington and Robinson, 1992; Taylor, 1997) was chosen as objective function:

$$\chi^2 = \sum_{j=1}^{N_p} \left[\bar{\Phi}_j^{\text{exp}} - \bar{\Phi}_j^{\text{sim}} \right]^2 \frac{1}{\sigma_j^2} \quad (22)$$

where $\bar{\Phi}_j^{\text{exp}}$ is the value of Φ measured in the experimental point “ j ”; $\bar{\Phi}_j^{\text{sim}}$ is the correspondent average value of Φ obtained by simulation; N_p is the number of experimental points and $1/\sigma_j^2$ is the statistical weight referring to the point “ j ”. If the statistical weights are unknown, they can be made equal to a common value, for instance 1. In Eq. (22), the chi-square depends on $\bar{\Phi}_j^{\text{sim}}$, which depends on I^{Φ} (i.e., parameters a_m or a_T and b_m or b_T) and h (h_m or h_T). Thus, the parameters a , b and h can be determined by minimization of Eq. (22) through successive trials.

3. Results and discussion

The circular section of the bananas was divided into 100 control volumes and the drying time was divided into 5000 steps. An unreported study on grid and time refinement indicated that such values are adequate to describe two processes: mass and heat transfer.

3.1. Results for water migration

The effective moisture diffusivity was considered variable as a function of the local moisture content as is given by Eq. (15), and the radius was also considered variable, according to Table 3. Thus, the parameters related with water transfer, obtained by optimization, are presented in Table 4.

The simulations of the drying kinetics are shown in Fig. 3a–c, and the superposition of these simulations is shown in Fig. 3d.

As additional information, the graphs of Fig. 3a–c were generated by own software developed to perform the optimizations.

Table 4
Parameters of the water migration obtained by optimization.

T ($^{\circ}\text{C}$)	a_m	b_m ($\text{m}^2 \text{min}^{-1}$)	h_m (m min^{-1})	R^2	χ^2
47.9	0.687	15.4×10^{-9}	2.66×10^{-5}	0.99970	15.6×10^{-3}
58.6	0.785	19.4×10^{-9}	2.68×10^{-5}	0.99989	3.87×10^{-3}
66.9	0.821	28.9×10^{-9}	2.99×10^{-5}	0.99964	9.20×10^{-3}

在本文中，用于将有效质量扩散率与局部水分含量联系起来的函数 $f(a, b, M)$ 可以表示为 (Silva 等人, 2012a)：

$$D = \frac{1}{4} \exp\left(\frac{aM}{b}\right) \quad (15)$$

其中， a 是需要通过优化来确定的参数，而 M 则表示干基下的局部水分含量。众所周知，产品的热扩散率取决于其成分。因此，新鲜水果的热扩散率非常高，因其含水量较高。在干燥过程中，水果在任何时刻的热扩散率都取决于其水分含量。随着干燥过程中水分的不断流失，即产品的水分含量逐渐减少，热扩散率的值也会随时间而降低 (Lima 等人, 2002 年; Mariani 等人, 2008 年)。因此，在本文中，对于每一种干燥空气温度的情况，都提出了以下表达式来描述这一特性：

$$a = \frac{1}{4} \exp\left(\frac{aM}{b}\right) \quad (16)$$

其中，各个干燥空气温度对应的常数是通过对优化计算得出的。另一方面， M 表示在干燥过程中任意时刻果实的平均水分含量。

2.4. 总体考虑因素 为了描述热传输过程，需要确定水流失率 dm/dt 。最初，人们先研究了水迁移现象。在每个时间步长中，都会使用三对角矩阵算法 TDMA (Press 等人, 1996 年) 来求解一组方程 (方程(4)、(6)和(8))。由于已经描述了水迁移过程，因此每个时间步长都会通过以下公式来计算水流失率： dm

$$\frac{dm}{dt} = \frac{1}{4} \frac{aM}{b} \quad (17)$$

在时刻 t 时，水的质量（单位：千克）可以表示为 $m(t) = m_0(t)$ 。其中， m_0 表示干物质的质量（单位：千克），该数值可在表 1 中找到；而 $m(t)$ 则代表以干基计算的平均水分含量（单位：千克每千克干物质），其计算方法如下：

$$M = \frac{1}{4} \frac{aM}{b} \quad (18)$$

其中， M_i 表示控制体积 “ i ” 中的水分含量， r_i 表示其半径（单位：米）。由于收缩效应，每个时间步长中圆柱体的半径都会发生变化，因此必须重新计算其数值。同时，均匀网格中控制体积的厚度 Δr 也需要重新确定。

这些计算的表达式在表 3 中给出。根据 Silva 等人 (2012b) 提供的公式，可以计算出香蕉表面水蒸气凝结的潜热。

$$h = \frac{1}{4} 2529:1M^2 2:386T \quad (19)$$

当 M 以干基单位给出， T 以摄氏度为单位时，他的数值用 kJ/kg 来表示。在给定的时间步长内，为了使用方程(4)、(6)和(11)来计算每个控制体积的温度，必须知道每个时间步开始和结束时的比热容和密度。这一点很明显，因为这些方程的系数 B 依赖于这些数值。香蕉的比热容是根据 Sweat 在 1986 年提出的农产品计算表达式来估算的： $c = 1381 \pm 2930X$ (20) 其中， X 表示平均水分含量，单位为湿基（小数形式）； c 则换算成 $\text{J/kg} \cdot \text{K}$ 单位后得出。

水蒸气的比热容 c 在表 1 中给出。另一方面，通过实验得到的测量数据，可以推导出香蕉的密度 ρ 与平均水分含量 M (db) 之间的关系式，该关系式是通过曲线拟合得到的：

$$\rho = \frac{1}{4} 1533e \quad (21)$$

因此，在每个时间步长中，如果热扩散系数和传热系数的数值由方程(4)、(6)和(11)给出，那么就可以求解该系统，从而确定每个控制体积中的温度。在每个时间步长，方程组都通过 TDMA 方法得到解。对于质量迁移的研究，同样适用类似的思路。需要注意的是，如果热扩散系数和传热系数未知，那么参数 a 和 h 可以通过优化方法（反方法）来确定，具体方法将在下文介绍。

2.5. 优化 为了确定参数 C (D 或) 的值，需要进行相应的优化计算。

a 和 h 函数 $C = f(a, b, U)$ 中的参数 a 和 b 可以通过使用实验数据集进行优化来确定。选择卡方分布作为目标函数表达式 (Bevington 和 Robinson, 1992 年; Taylor, 1997 年)。

$$\chi^2 = \sum_{j=1}^N \frac{(U_j - U_{\text{model}})^2}{U_{\text{model}}} \quad (22)$$

其中， U_j 是在实验点 “ j ” 处测得的 U 值； U_{model} 则是通过模拟得到的相应 U 值的平均值； N 是实验点的数量， $l =$

$\frac{1}{r}$ 统计权重指的是点 “ j ” 的权重。如果统计权重未知，可以将它们设置为一个共同的值，例如 1。在方程(22)中，卡方分布取决于 U ，而 U 又取决于 C (即参数 a 或 h)，以及 b (即 U 或 h)。因此，可以通过最小化方程来确定参数 a 、 b 和 h 的值。你的输入是否正确，或者提供更多的上下文信息。

(22) 通过不断的试验和尝试。

3. 结果与讨论 香蕉的圆形截面被划分为 100 个控制体积，而干燥时间则被划分为 5000 个阶段。一项关于网格和时间精细化的未公开研究表明，这些划分方式足以描述两个过程：物质传递与热量传递。

3.1. 水迁移的结果 有效水分扩散系数被视为随局部水分含量而变化的函数，具体表达式如方程(15)所示。同时，半径也被视为可变的参数，具体数值根据表 3 进行确定。因此，通过优化得到的与水迁移相关的参数在表 4 中有所呈现。干燥动力学的模拟结果如图 3a–c 所示，而这些模拟结果的叠加则如图 3d 所示。作为补充信息，图 3a 至图 3c 中的图表是由我们自行开发的软件生成的，该软件专门用于执行优化计算。

表 4
通过优化得到的水迁移参数。

温度 (°C)	a	b (mm)	h (分钟)	R	v
47.9	0.687	15.4 10	2.66 10	0.99970	15.6 10
58.6	0.785	19.4 10	2.68 10	0.99989	3.87 10
66.9	0.821	28.9 10	2.99 10	0.99964	9.20 10

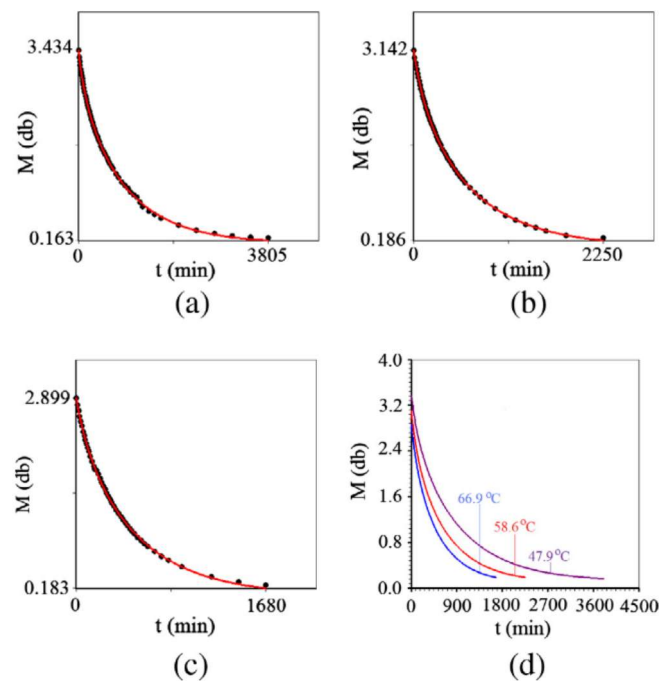


Fig. 3. Simulation of water migration at drying air temperature of: (a) 47.9 °C; (b) 58.6 °C; (c) 66.9 °C; (d) superposition of the simulated curves.

On the other hand, with the results from Table 4, the graphs of the functions $D(M)$, given by Eq. (15), are presented in Fig. 4 for three temperatures of the drying air.

In order to present the moisture distributions within the circular section that represents the bananas, Fig. 5 highlights these distributions at 200 min, for three temperatures of the drying air.

3.2. Results for heat transport

Thermal diffusivity was considered variable as a function of the average moisture content, as is given by Eq. (16). In the heat transport study, the radius was also considered variable, according to Table 3. Thus, the parameters related with the heat transfer, obtained by optimization, are presented in Table 5.

The simulations of the heating kinetics in the center of the bananas are shown in Fig. 6a–c, while Fig. 6d presents the superposition of these simulations.

Note that the graphs of Fig. 6a–c were also generated by software developed to perform the optimization processes.

With the results from Table 5, the graphs of the functions $\alpha(\bar{M})$, given by Eq. (16), are presented in Fig. 7, for three temperatures of the drying air.

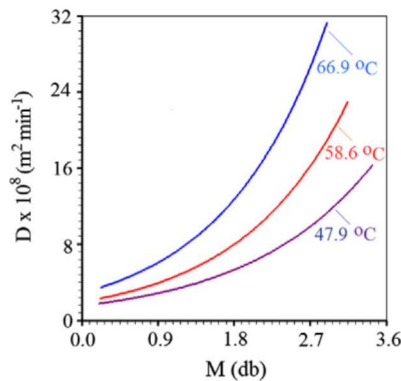


Fig. 4. Effective moisture diffusivity as a function of the local moisture content at temperature of: (a) 47.9 °C; (b) 58.6 °C; (c) 66.9 °C.

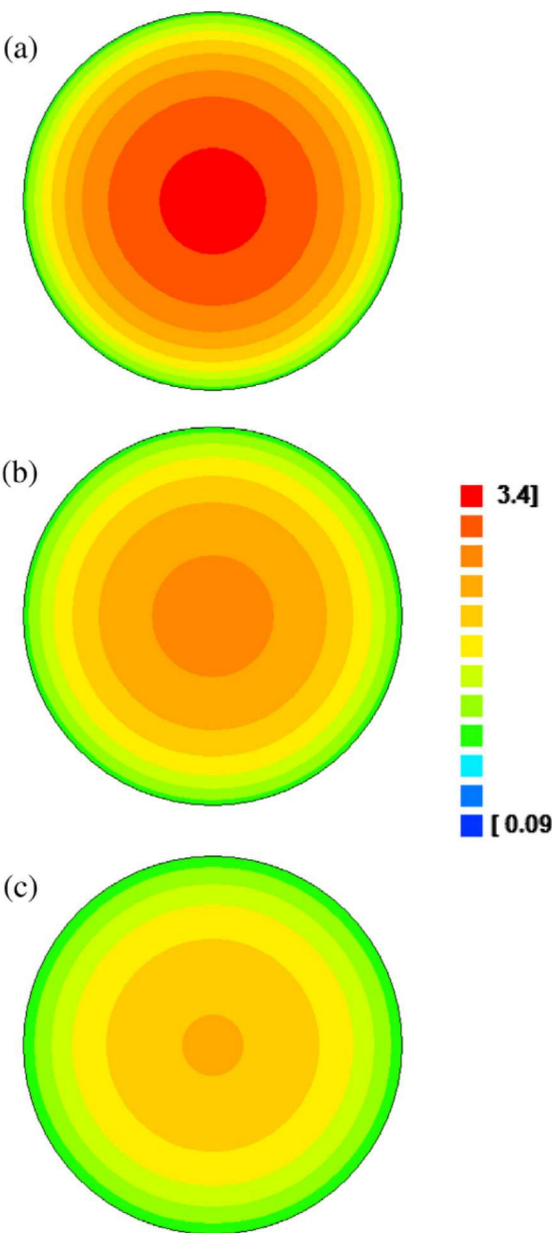


Fig. 5. Distribution of the moisture content at $t = 200$ min for the drying air temperature of: (a) 47.9 °C; (b) 58.6 °C; (c) 66.9 °C.

Table 5
Parameters of the heat conduction obtained by optimization.

T (°C)	a_T	b_T (m ² min ⁻¹)	h_T (m min ⁻¹)	R^2	χ^2
47.9	0.313	1.26×10^{-7}	5.61×10^{-4}	0.99071	14.3
58.6	0.398	1.06×10^{-7}	5.71×10^{-4}	0.98843	28.9
66.9	0.520	1.26×10^{-7}	9.49×10^{-4}	0.98084	33.4

The distributions of temperature within the circular section that represents the bananas at $t = 10$ min are given in Fig. 8 for three temperatures of the drying air.

3.3. Analysis of simplifications for the heat transfer problem

3.3.1. Exclusion of the heating of the vapor

In the proposed model, vapor is generated at temperature of the surface and is heated to the drying air temperature. In order to test the influence of the heating of the vapor, a simulation was

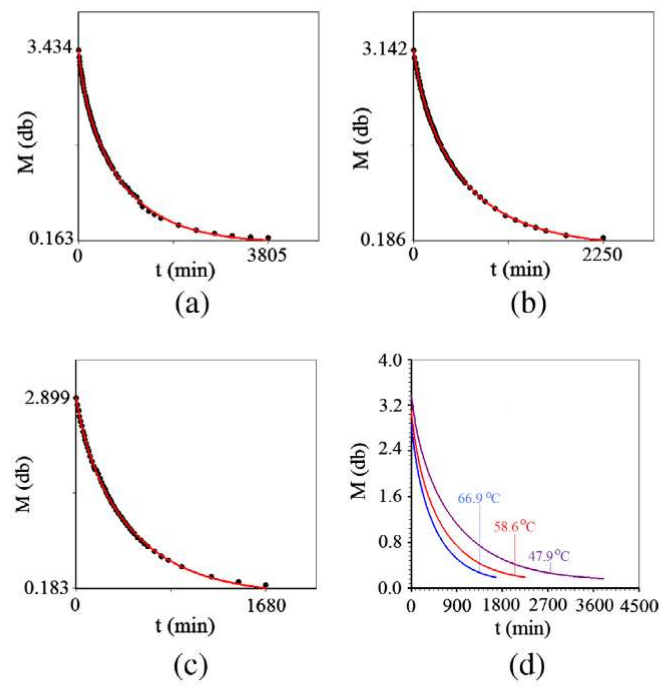


图 3. 在不同空气温度下的水分迁移模拟结果: (a) 47.9°C; (b) 58.6°C; (c) 66.9°C; (d) 各种模拟曲线的叠加图。

另一方面，根据表 4 的结果，函数 $D(M)$ 的图像在图 4 中呈现出来。这些图像对应于三种不同干燥空气的温度条件。

为了展示代表香蕉的圆形截面内的水分分布情况，图 5 展示了在 200 分钟时，三种不同干燥空气温度下的水分分布情况。

3.2. 热传输的结果：热扩散率被视为随平均水分含量变化的变量，具体如方程(16)所示。在热传输研究中，半径也被视为可变的参数，具体数据可参见表 3。因此，通过优化得到的与热传递相关的参数，在表 5 中有所呈现。

香蕉中心加热过程的模拟结果如图 6a–c 所示，而图 6d 则展示了这些模拟结果的叠加情况。

请注意，图 6a 至图 6c 中的图形也是通过专门开发的软件生成的，该软件用于执行优化过程。

根据表 5 的结果，方程(16)所给出的函数图形在图 7 中得到了展示，这些图形对应于三种不同的干燥空气温度情况。

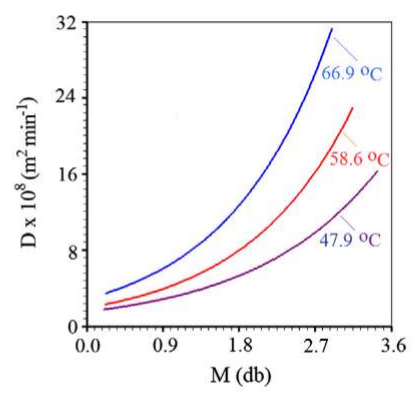


图 4. 不同温度下，局部水分含量时的有效水分扩散系数函数: (a) 47.9°C; (b) 58.6°C; (c) 66.9°C。

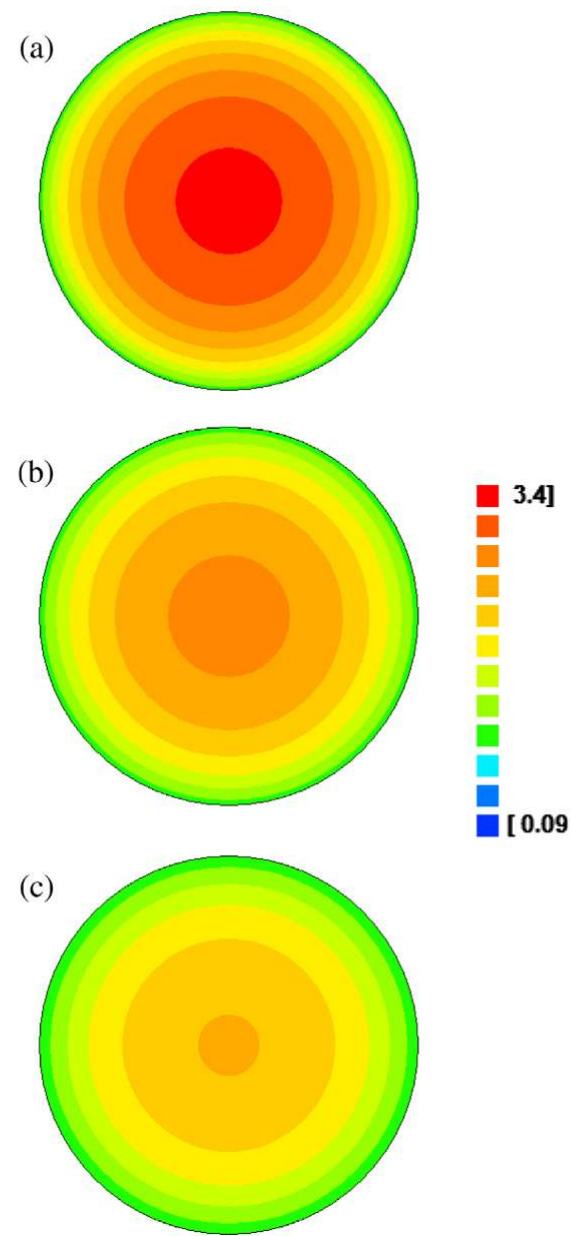


图 5. 在 $t=200$ 分钟时，不同干燥空气温度下的水分分布情况: (a) 47.9°C; (b) 58.6°C; (c) 66.9°C。

表 5
通过优化得到的热传导参数。

温度 (°C)	a	b (m 最小值)	h (分钟)	R	v
47.9	0.313	1.26 10	5.61 10	0.99071	14.3
58.6	0.398	1.06 10	5.71 10	0.98843	28.9
66.9	0.520	1.26 10	9.49 10	0.98084	33.4

在图 8 中，展示了在 $t=10$ 分钟时，圆形区域内的温度分布情况。图中展示了三种不同干燥空气温度下的香蕉温度情况。

3.3. 热传递问题简化方法的分析

3.3.1. 蒸汽加热的排除在所提出的模型中，蒸汽是在表面温度下产生的，然后被加热到与干燥空气相同的温度。为了测试蒸汽加热的影响，进行了一项模拟研究。

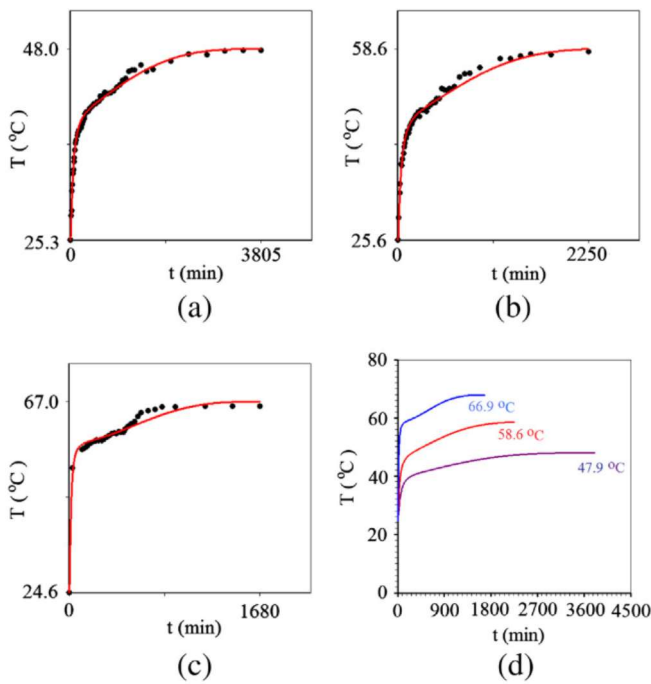


Fig. 6. Temperature in the center of the banana at drying air temperature of: (a) 47.9 °C; (b) 58.6 °C; (c) 66.9 °C; (d) superposition of the simulated curves.

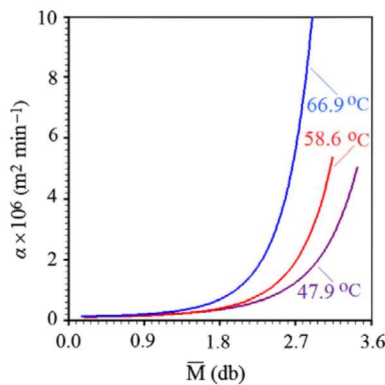


Fig. 7. Thermal diffusivity as a function of the average moisture content at drying air temperature of: (a) 47.9 °C; (b) 58.6 °C; (c) 66.9 °C.

performed at 58.6 °C, considering the parameters given in Table 5, but supposing $c_v = 0$. In this case, the statistical indicators of the simulation were $\chi^2 = 29.2$ (instead 28.9) and $R^2 = 0.98835$ (instead 0.98843). A graph showing the simulation of the heating kinetics at the center of the banana does not present a significant visual difference when compared with Fig. 6b and, therefore, was not presented herein. Similar results were obtained for the drying air temperatures of 47.9 and 66.9 °C.

3.3.2. Latent heat of water vaporization

In a simulation with the drying air temperature at 58.6 °C, if the results obtained in Table 5 are used, but latent heat of water vaporization is neglected ($h_{fg} = 0$), the statistical indicators are: $\chi^2 = 4146.1$ (instead 28.9) and $R^2 = 0.75502$ (instead 0.98843). These results are completely unacceptable, and similar results are obtained for the drying air temperatures of 47.9 and 66.9 °C. On the other hand, at 58.6 °C, the use of the value of the latent heat of vaporization of free water in the simulation resulted in statistical indicators given by $\chi^2 = 29.6$ (instead 28.9) and $R^2 = 0.98832$ (instead 0.98843). Similar results were obtained at 47.9 and 66.9 °C.

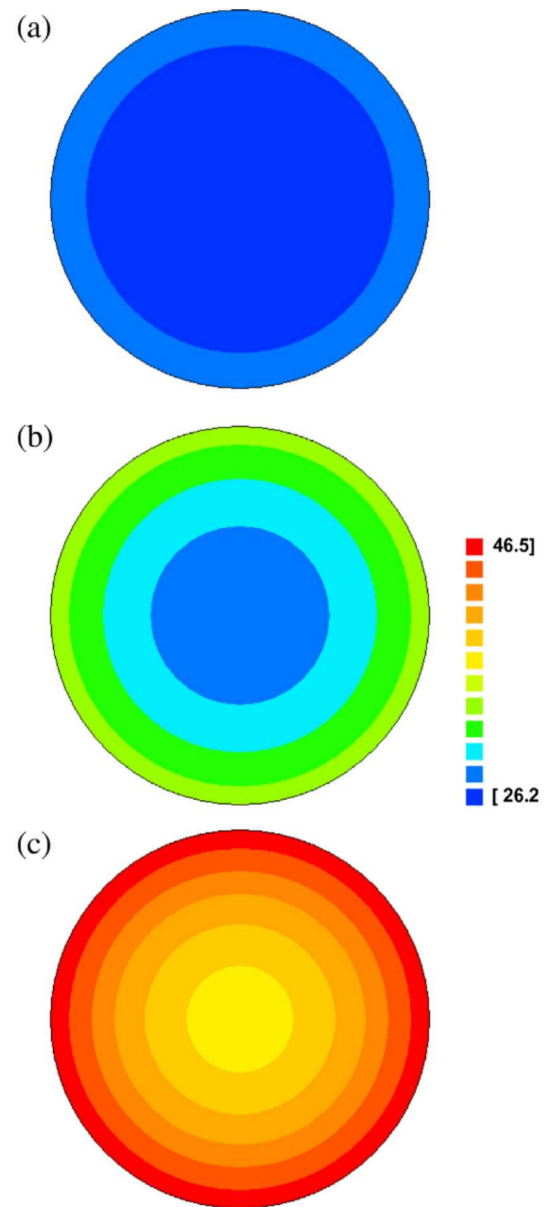


Fig. 8. Distribution of the temperatures at $t = 10$ min for the drying air temperature of: (a) 47.9 °C; (b) 58.6 °C; (c) 66.9 °C.

3.3.3. Constant density and specific heat of bananas

In many occasions, the transient term of Eq. (1) applied to heat transfer, given by $\partial(\rho c_p T)/\partial t$, is written as $\rho c_p \partial T/\partial t$, i.e., in this term the product ρc_p is considered constant. In order to test this simplification for the drying case, the results obtained in Table 5 for $T = 58.6$ °C were used in a simulation, considering the transient term given by $\rho c_p \partial T/\partial t$. As result, the chi-square was 264.1 (instead 28.9) and the determination coefficient was 0.97835 (instead 0.98843). Similar results were also obtained for the drying air temperatures of 47.9 and 66.9 °C, indicating that this simplification should be avoided in drying studies.

3.4. Discussion

Basically, in the transient diffusion study, two methods are used to determine the thermal diffusivity and heat transfer coefficient of agricultural products: empirical correlations (Karim and Hawlader, 2005; Perussello et al., 2013) and optimization (Lima et al., 2002;

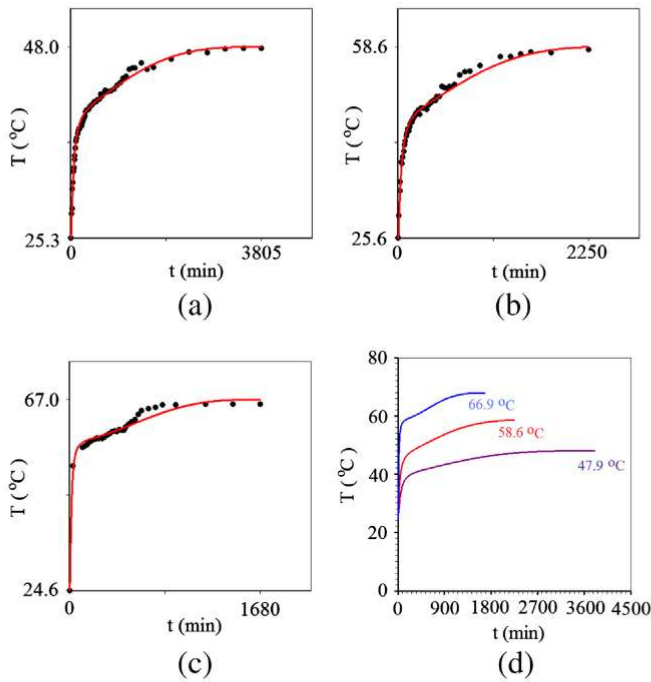


图 6. 在干燥空气温度下, 香蕉中心的温度情况: (a) 47.9°C; (b) 58.6°C; (c) 66.9°C; (d) 各种模拟曲线的叠加图。

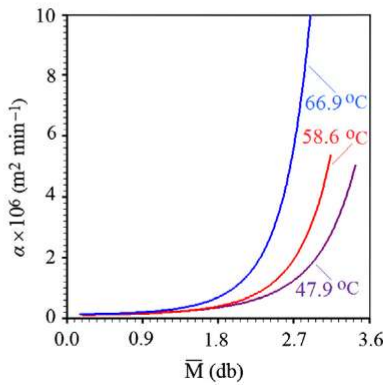


图 7. 在不同干燥空气温度下的平均水分含量时, 热扩散系数的变化情况: (a) 47.9°C; (b) 58.6°C; (c) 66.9°C。

在 58.6°C 的温度下进行了实验, 参考了表 5 中的参数, 但假设 $c=0$ 。在这种情况下, 模拟的统计指标为 $v=29.2$ (而非 28.9), 而 $=0.98835$ (而非 0.98843)。在香蕉中心处进行的加热动力学模拟结果, 与图 6b 相比没有明显的视觉差异, 因此这里不再展示。对于干燥空气温度分别为 47.9°C 和 66.9°C 的情况, 也得到了类似的结果。

3.3.2 水汽化的潜热 在模拟中, 干燥空气的温度为 58.6°C, 如果采用表 5 中的结果, 但忽略水汽化的潜热 (即 $h=0$), 那么统计指标如下:

V

这些结果完全不可接受。在干燥空气温度为 47.9°C 和 66.9°C 的情况下, 也得到了类似的结果。另一方面, 当干燥空气温度为 58.6°C 时, 如果使用自由水的汽化热值来进行模拟计算, 那么统计指标将分别变为 $v=29.6$ (而不是 28.9), 以及 (而不是 0.98843)。在 47.9°C 的情况下, 也得到了类似的结果。

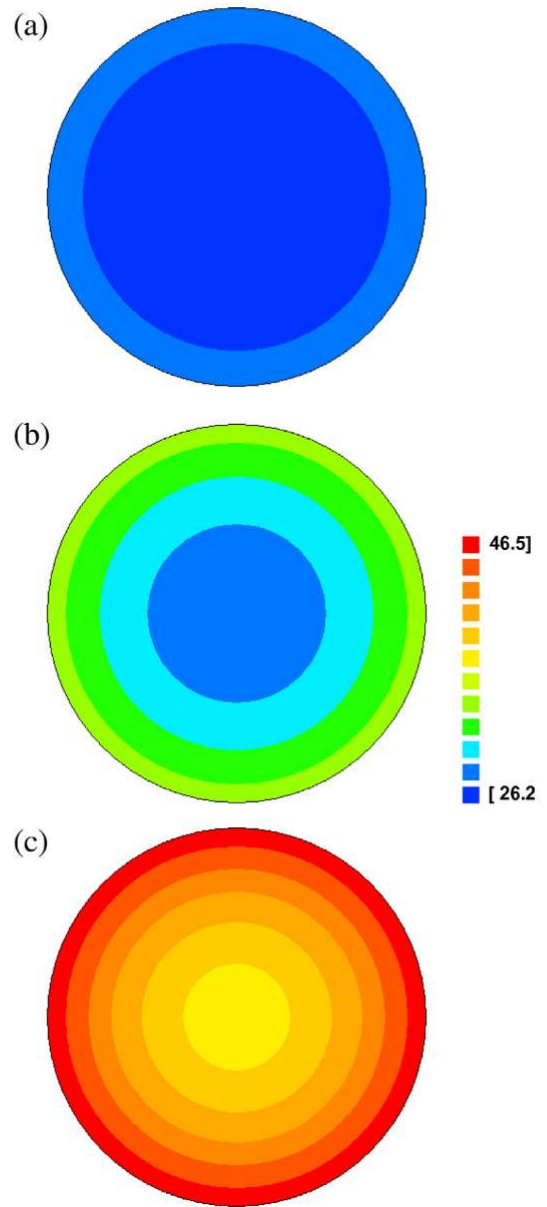


图 8. 在 $t=10$ 分钟时, 不同干燥空气温度的分布情况: (a) 47.9°C; (b) 58.6°C; (c) 66.9°C。

3.3.3 香蕉的恒定密度和比热容 在许多情况下, 方程(1)中的瞬态项用于描述热传递过程, 其表达式为 $@(q,)$ 。在这里, 乘积 q 被视为常数。为了验证这种简化在干燥过程中的适用性, 我们采用了表 5 中在 $T=58.6^\circ\text{C}$ 时的计算结果。在模拟过程中, 将瞬态项替换为 q (而不是 28.9), 同时确定系数被设定为 0.97835 (而不是 0.98843)。在干燥空气温度为 47.9°C 和 66.9°C 的情况下, 也得到了类似的结果。这表明在干燥研究过程中应该避免这种简化方式。

3.4. 讨论

在瞬变扩散研究中, 为了确定农产品的热扩散系数和传热系数, 通常采用两种方法来获取这些数据: 经验相关性分析 (Karim 和 Hawlader, 2005 年; Perussello 等人, 2013 年), 以及优化方法 (Lima 等人, 2002 年)。

Silva et al., 2011, 2013). In the present article, only optimizations were used to determine α and h_T , and the obtained results are compatible with those reported in the literature for bananas (Lima et al., 2002; Mariani et al., 2008).

Trough an inspection of Table 4 it is possible to affirm that the statistical indicators to describe the mass transfer are excellent. Particularly, the determination coefficients are greater than 0.99960. Similar conclusion on the adequacy of the proposed model can be obtained by an observation of Fig. 3a–c for three temperatures of the drying air. On the other hand, Fig. 3d presents a superposition of the drying kinetics, in which it is possible to evaluate how much one simulation is faster than other. From Table 4 and Eq. (15), the minimum and maximum values of the effective moisture diffusivity are $2.79 \times 10^{-10} \text{ m}^2 \text{ s}^{-1}$ (47.9 °C and $M = 0.121$ db) and $5.21 \times 10^{-9} \text{ m}^2 \text{ s}^{-1}$ (66.9 °C and $M = 2.90$ db). Details on effective moisture diffusivity as a function of the local moisture content are presented in Fig. 4. In this figure it is realized that, the more the temperature of the drying air is high, the greater the effective moisture diffusivity. This obtained result is a consequence of the following fact: increasing the drying air temperature increases the overall product temperature and, hence, the moisture diffusivity. On the other hand, the convective mass transfer coefficient varied from 4.43×10^{-7} to $4.98 \times 10^{-7} \text{ m s}^{-1}$. Similar results were obtained by Silva et al. (2012a) and Lima et al. (2002) studying drying of bananas.

The moisture distributions are presented in Fig. 5, with no scale, for three drying air temperatures, at $t = 200$ min. An observation of this figure allows to conclude that the more the temperature of the drying air is high, the greater the loss of moisture. Although this conclusion is quite obvious, the calculations performed herein are necessary for a subsequent determining the stress within the fruit due to the difference in the moisture concentration.

Table 5 presents the results of the optimizations for the temperatures measured in the center of the bananas. The statistical indicators show that the obtained results are reasonable. Particularly, the determination coefficients are greater than 0.98000. A visual observation of Fig. 6a–c makes it possible to conclude that the simulated lines agree with experimental datasets. On the other hand, Fig. 6d enables to realize that the heating process is very fast. Differently of water migration, few minutes are enough so that the temperature at center of the fruit approximates of its equilibrium value. This fact permitted to establish the assumption number 3 of this article: “the mass migration is considered under isothermal conditions, since at initial instants of drying the thermal diffusivity ($10^{-7} \text{ m}^2 \text{ s}^{-1}$) is greater than the moisture diffusivity ($10^{-10} \text{ m}^2 \text{ s}^{-1}$)”.

Inspection of Fig. 7 indicates that, for all temperatures of the drying air, the thermal diffusivity exponentially decreases with the decreasing of average moisture content of the banana. Shyamal et al. (1994), studying the thermal properties of the wheat, concluded that the thermal diffusivity decrease linearly with moisture content. However, the range of the moisture content in drying of grains is lesser than the range of fruits such as bananas. A similar result to the obtained herein for thermal diffusivity of bananas was obtained by Mariani et al. (2008). However, the expressions proposed by these researchers for the thermal diffusivity involved three and four parameters and, in the present article, only two parameters were involved. In the present article, at the beginning of drying, the values for the thermal diffusivities are given by 0.835×10^{-7} (47.9 °C), 0.894×10^{-7} (58.6 °C) and $1.67 \times 10^{-7} \text{ m}^2 \text{ s}^{-1}$ (66.9 °C). Based on the literature, these results are acceptable for agricultural products with high moisture content, such as bananas. At the end of the drying processes, the values obtained in this work for thermal diffusivity are of about $2.0 \times 10^{-9} \text{ m}^2 \text{ s}^{-1}$. The highest value obtained in the present paper for the thermal diffusivity ($1.67 \times 10^{-7} \text{ m}^2 \text{ s}^{-1}$) is lower than those obtained by Mariani et al. (2008), 1.88×10^{-7} , and Lima et al.

(2002), $2.70 \times 10^{-7} \text{ m}^2 \text{ s}^{-1}$. On the other hand, the value obtained herein is closest to the value of the thermal diffusivity of the water, as is expected. In addition, the lower value obtained in this article, at the end of drying, is compatible with two works above-mentioned.

In their article, Mariani et al. (2008) have determined the values for convective heat transfer coefficient based on the Nusselt number, i.e., using empirical correlations instead optimization. Their results were from 15 to $35 \text{ W m}^{-2} \text{ °C}^{-1}$. In the present work, this property was determined by optimization and, at beginning of the process, ranged between 15.2 (47.9 °C) and $28.3 \text{ W m}^{-2} \text{ °C}^{-1}$ (66.9 °C). These results are compatible with those from Mariani et al. (2008), and also agree with those obtained by Lima et al. (2002): from 12.7 to $19.0 \text{ W m}^{-2} \text{ °C}^{-1}$.

From Fig. 8 it is possible to realize that, at instant $t = 10$ min, the range of temperature in the interior of the banana increases with the increasing of the drying air temperature. Remembering that bananas are at room temperature at the initial instant, at drying air temperature of 47.9 °C and $t = 10$ min the range of temperature in the circular section was from 26.2 (center) to 28.9 °C (surface). For the drying air temperatures of 58.6 and 66.9 °C, these ranges were 28.4–34.0 and 37.3–46.4 °C, respectively. Finally, although the three circles of Fig. 8 have the same size, i.e., three figures are drawn with no scale, it is expected that the circle related to drying at 66.9 °C has lesser radius than other circles, once the banana at this temperature dries faster than the other two cases. Obviously, the shrinkage due to water loss is more important than the effect of expansion by temperature increase.

Lima et al. (2002) and Mariani et al. (2008), in their analyses, included the specific heat of the water vapor in order to describe heat transport. However, several researchers such as Rinaldi et al. (2011) and also Ramsaroop and Persad (2012), studying simultaneous heat and mass transfer, did not consider this phenomenon. The simulations performed in the present work made it possible to realize that the heating of vapor at surface temperature to drying air temperature is not significant in the description of the heat transport during convective drying of bananas. On the other hand, Wu et al. (2004) studied heat and mass transfer during drying of rice. These researchers used two diffusion equations to describe water and heat migration, supposing that two phenomena are independent, i.e., the latent heat of water vaporization was neglected. However, the results obtained in the present article indicated that the coupling between two phenomena is very important and the latent heat of water vaporization cannot be neglected. Another simulation performed in the present article has shown that the use of the latent heat of vaporization of free water (instead latent heat of vaporization of water in the product) is not significant to describe drying of bananas.

Many times the product ρc_p in the transient term is considered constant in the equation of heat conduction (Mariani et al., 2008; Ramsaroop and Persad, 2012; Lemus-Mondaca et al., 2013). This consideration made it possible to determine an analytical solution for the equation of heat conduction in a hemispherical shell geometry (Ramsaroop and Persad, 2012). However, the results obtained in the present paper indicated that such consideration worsened the results of model proposed and, consequently, should be avoided.

4. Conclusion

Liquid diffusion model, with vaporization of water at surface of the bananas, was assumed in this article. Model included mass and thermal diffusivities as variable properties, boundary conditions of the third kind, and also included the shrinkage. Using an infinite cylinder to represent the bananas, this model well described the mass and heat transfer during the drying process.

Silva 等人, 2011 年、2013 年)。在本文中, 仅使用了优化方法来确定该参数的值, 所获得的结果与文献中关于香蕉的研究结果一致 (Lima 等人, 2002 年; Mariani 等人, 2008 年)。

通过查看表 4, 可以确认用于描述质量传递的统计指标非常优秀。特别是, 决定系数都大于 0.99960。通过观察图 3a 至 c 中三种干燥空气温度下的数据, 也可以得出类似的结论。另一方面, 图 3d 展示了干燥动力学的叠加情况, 从中可以评估某个模拟模型比其他模型快多少。根据表 4 和方程(15), 有效水分扩散率的最小值和最大值分别为 2.79×10^{-3} ms(47.9°C 时, $M=0.121$ db)和 5.21×10^{-3} ms(66.9°C 时, $M=2.90$ db)。关于有效水分扩散率与局部水分含量关系的详细信息可以在图 4 中找到。从图中可以看出, 干燥空气的温度越高, 有效水分扩散率就越大。这一结果源于这样一个事实: 提高干燥空气的温度会提高产品的整体温度, 从而提升水分的扩散能力。另一方面, 对流传质系数在 4.43×10^{-3} 到 4.98×10^{-3} 米每秒之间变化。Silva 等人 (2012a) 以及 Lima 等人 (2002) 在研究香蕉的干燥过程时也得到了类似的结果。

图 5 展示了在 $t=200$ 分钟时, 三种干燥空气温度下的水分分布情况。从图中可以观察到, 干燥空气的温度越高, 水分的流失就越严重。虽然这一结论显而易见, 但在此进行的计算对于后续确定果实内部因水分浓度差异而产生的应力是必不可少的。

表 5 展示了对香蕉中心温度进行优化后的结果。统计指标表明, 所获得的结果相当合理。具体而言, 决定系数均超过 0.98000。通过图 6a 至图 6c 的直观观察可以得出结论: 模拟曲线与实验数据一致。另一方面, 图 6d 显示加热过程非常迅速。与水分迁移不同, 只需几分钟即可使果实中心的温度接近其平衡值。这一事实支持了本文中的第三个假设: “在等温条件下可以考虑质量迁移现象, 因为在干燥初期, 热扩散率 (10 毫秒) 大于水分扩散率 (10 毫秒)”。

对图 7 的观测表明, 在所有的干燥空气温度下, 随着香蕉平均水分含量的降低, 热扩散系数呈指数级下降。Shyamal 等人 (1994 年) 在研究小麦的热性质时发现, 热扩散系数与水分含量呈线性下降关系。然而, 谷物干燥过程中水分含量的变化范围小于香蕉等水果的情况。Mariani 等人 (2008 年) 也获得了类似的结果。不过, 这些研究者提出的热扩散系数表达式涉及三个或四个参数, 而本文所采用的模型仅涉及两个参数。在本文中, 干燥初期热扩散系数的数值分别为 0.835×10^{-1} (47.9°C)、 0.894×10^{-1} (58.6°C) 和 1.67×10^{-1} (66.9°C)。根据文献资料, 这些结果对于水分含量较高的农产品来说是可以接受的, 比如香蕉。在干燥过程结束时, 本研究得到的热扩散系数约为 2.0×10^{-3} 米每秒。本文中获得的热扩散系数最大值 (1.67×10^{-3} 米每秒) 低于 Mariani 等人 (2008 年) 的研究结果, 后者的值为 1.88×10^{-3} 米每秒。

(2002 年), 2.70×10^{-3} 毫秒。另一方面, 本文中得到的数值最接近于水的热扩散率, 这与预期一致。此外, 本文在干燥过程结束时得到的较低数值与上述两项研究结果也是相一致的。

在他们的文章中, Mariani 等人 (2008 年) 基于努塞尔数来确定了对流传热系数的数值, 即采用经验关系而非优化方法。他们的计算结果范围为 15 到 35 瓦特每摄氏度。在本研究中, 该属性是通过优化方法确定的, 在过程开始时, 其数值介于 15.2 瓦特每摄氏度 (47.9 摄氏度) 到 28.3 瓦特每摄氏度 (66.9 摄氏度) 之间。这些结果与 Mariani 等人 (2008 年) 的研究结果一致, 同时也与 Lima 等人获得的结果相吻合。

(2002 年): 温度从 12.7 摄氏度升至 19.0 摄氏度。

从图 8 可以看出, 在 $t=10$ 分钟时, 随着干燥空气温度的升高, 香蕉内部的温度范围也在扩大。需要注意的是, 在初始状态下, 香蕉的温度处于室温; 当干燥空气的温度达到 47.9°C 时, 经过 10 分钟后, 圆形截面的温度范围从 26.2°C (中心) 上升到 28.9°C (表面)。当干燥空气的温度分别达到 58.6°C 和 66.9°C 时, 相应的温度范围分别变为 28.4°C–34.0°C 和 37.3°C–46.4°C。虽然图 8 中的三个圆圈大小相同, 即三个图形都是无比例尺绘制的, 但预计在 66.9°C 的温度下, 圆圈的半径会比其他两个情况下的圆圈更短, 因为此时香蕉的干燥速度会比其他两种情况更快。显然, 由于水分流失导致的收缩效应, 比温度升高引起的膨胀效应更为重要。

Lima 等人 (2002 年) 和 Mariani 等人 (2008 年) 在分析中考虑了水蒸气的比热容, 以描述热量传输过程。然而, 一些研究人员如 Rinaldi 等人 (2011 年) 以及 Ramsaroop 和 Persad (2012 年) 在研究同时发生的热量和质量传递时, 并未考虑这一现象。本研究的模拟结果表明, 在香蕉的蒸发过程中, 水蒸气在表面温度到干燥空气温度之间的升温效应对热量传输的描述并不重要。另一方面, Wu 等人 (2004 年) 研究了大米干燥过程中的热量和质量传递现象。这些研究人员使用了两个扩散方程来描述水分和热量的迁移, 并假设这两种现象是相互独立的, 即忽略了水蒸发的潜热效应。不过, 本文的研究结果表明, 这两种现象之间确实存在耦合关系。蒸发潜热非常重要, 因此不能忽视它。本文中进行的一项模拟表明, 使用自由水的蒸发潜热 (而非产品中的水的蒸发潜热) 来描述香蕉的干燥过程并不具有显著效果。

在许多情况下, 热传导方程中的瞬变项被视作常数 (Mariani 等人, 2008 年; Ramsaroop 和 Persad, 2012 年; Lemus-Mondaca 等人, 2013 年)。这种处理方式使得在半球形壳体的几何结构下能够求得热传导方程的解析解

(Ramsaroop 和 Persad, 2012 年)。然而, 本文的研究结果表明, 这种处理方式会恶化所提出的模型的预测效果, 因此应避免使用这种方法。

4. 结论

在本文中, 采用了液体扩散模型来描述香蕉的干燥过程。该模型将质量传递和热传导视为可变属性来处理; 同时考虑了边界条件中的第三种情况, 以及香蕉的收缩现象。通过使用无限大的圆柱体来表示香蕉, 该模型能够很好地描述干燥过程中物质的质量传递和热量传递现象。

In the description of mass migration, the determination coefficients were greater than 0.99960 for all drying air temperatures. Also, it can be concluded that the expressions obtained for $D(M)$, as well as the values for convective mass transfer coefficients are adequate to describe mass migration during drying of bananas. On the other hand, in the description of heat transport, the determination coefficients were greater than 0.98000. In addition, the values obtained for thermal diffusivity and convective heat transfer coefficient are compatible with the expected values for mature bananas submitted to the experiments performed. Thus, it can be concluded that the expressions obtained for $\alpha(\bar{M})$ as well as the values for convective heat transfer coefficients are adequate to describe heat transport during drying.

It was possible to realize that the exclusion of the energy used to heat the vapor from the surface temperature up to the drying air temperature does not influence the description of the heating of bananas in a significant way. However, the exclusion of latent heat of vaporization strongly worsened the obtained results. On the other hand, the use of the value of the latent heat of vaporization of free water in the simulations (instead latent heat of vaporization of water in the product) is not significant to describe the process. Finally, it was concluded that the consideration of the product ρc_p with a constant value in the equation of heat conduction should be avoided.

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在描述物质迁移的过程中，所有干燥空气温度下的确定系数均大于 0.99960。此外，可以得出结论：用于描述香蕉干燥过程中物质迁移的表达式以及对流质量传递系数的数值都是合适的。另一方面，在热量传输的描述中，确定系数大于 0.98000。另外，所得到的热扩散系数和对流热传递系数的数值与实验中预期的成熟香蕉的相应数值是一致的。因此，可以得出结论：用于描述香蕉干燥过程中物质迁移的表达式是有效的。

a

这些对流传热系数的数值足以描述干燥过程中的热量传递情况。

可以观察到，将用于加热蒸汽的能源从表面温度到干燥空气温度这一过程中排除不干，并不会显著影响香蕉的加热过程的描述。然而，如果排除汽化潜热，那么得到的计算结果就会严重恶化。另一方面，在模拟中采用自由水的汽化潜热值（而非产品中的水的汽化潜热值）来描述这一过程也是无效的。最终得出结论：在热传导方程中，应避免将产品 q 视为一个恒定值。

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